Assignment # 3

Artificial Intelligence

Total Marks: 80 **Submission Date: 19-05-2026**

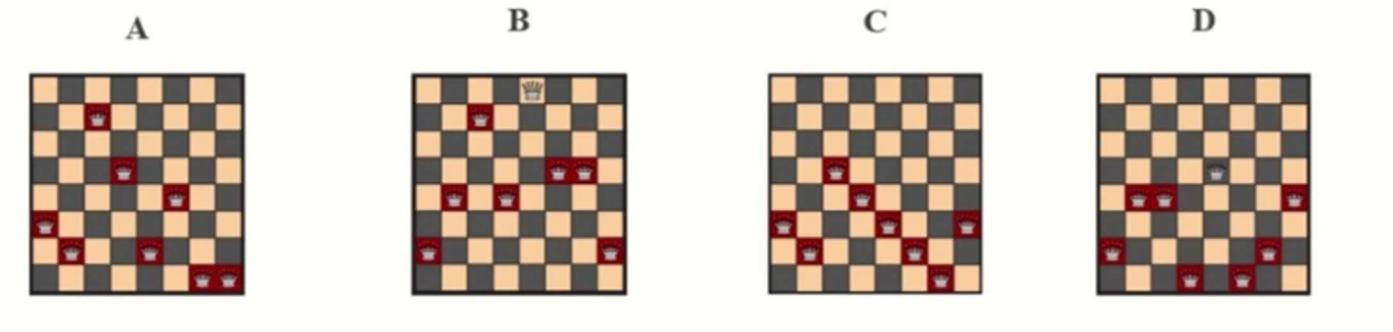
Name_____**Roll No.**_____

Instructions:

I. Assignment should be hand written. It will be submitted in class while its soft copy is also submitted on portal.

II. You should create your own solution according to your logic. Any similarity with other student and internet will be consider as cheating and your assignment will be marked as ZERO.

Question #1. **20 Marks**



1. Evaluate fitness function of each state (no of non-attacking pairs)
   - Selection method of two pairs (after arranging in descending order)
   - Two-point Crossover operation after 4th and 7th Queen
   - Mutation operation by swapping 1st and 4th Queen
2. After this create new generation and Tell either the next generation is better or not

Question #2. **20Marks**

You are given a knapsack with a maximum weight capacity of **7 pounds**. There are **4 items**, each with a specific weight and monetary value:

Item	Weight	Value
1	2 lb	$3.00
2	3 lb	$2.00
3	4 lb	$4.00
4	5 lb	$1.00

Initial population

Chromosome
C1: 1100
C2: 1010
C3: 0110
C4: 1001
C5: 0101
C6: 0011

Tasks:

1. Represent potential solutions using a binary chromosome, where each gene (bit) indicates whether a corresponding item is included (1) or excluded (0).
2. Define the fitness function based on the total value of selected items, ensuring the total weight does not exceed the knapsack's limit.
3. Implement the core components of a GA:
    - Selection method (roulette wheel)
    - Two-point Crossover operation after 1st and 3rd gene
    - Mutation operation by swapping 1st and 4th gene
4. Apply your algorithm to the given data and report:
    - Tell either the next generation is better or not
    - The best solution found (set of items)
    - Its total weight
    - Its total value

Question #3 **20 Marks**

A FedEx van starts its delivery route from the European hub in **Brussels**, where it must collect letters and packages from the regional collection centers located in **London**, **Berlin**, **Madrid**, and **Paris**. The van must:

- Visit **each city exactly once**,
- Then return to **Brussels**,
- And travel the **shortest total distance** possible.

You are given the **distance matrix** between all cities (as shown below). Your task is to solve this problem using a **Genetic Algorithm** approach.

	London	Berlin	Madrid	Brussels	Paris
London	0	1059	1704	328	414
Berlin	1059	0	2527	655	1094
Madrid	1704	2527	0	1562	1262
Brussels	328	655	1562	0	294
Paris	414	1094	1268	294	0

Tasks:

1. Design a **Genetic Algorithm** to find a near-optimal route for the TSP described above. (Population size should be > 4)
2. Define the representation of individuals, the fitness function, selection, crossover (2 point after 1st and 3rd city), and mutation (swapping 2nd and 3rd position) strategies.

Question #4 **20 Marks**

Apply alpha beta pruning on the following tree by visiting it from left to right and highlight the pruned branches also mention either alpha or beta cut applied.

Note: symbols of max and Min also shown

